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# Executive Summary

The U.S. single-family and low-rise residential sector remains overwhelmingly wood-framed, while mid-rise/multifamily (<9 stories) is rapidly adopting engineered timber (CLT/glulam) and modular systems, and high-rise/office towers rely on steel and concrete. **Material costs have surged since 2020** – lumber, steel and concrete remain 40–60% above pre-COVID levels【17†L135-L143】. This has raised the share of materials to ~50% of total construction cost【17†L139-L146】. Key cost drivers include commodity prices (e.g. iron ore, cement), energy/fuel (cement kilns, EAF steelmaking), labor shortages (US needs ~500k new construction workers in 2026【52†L113-L122】), tariffs and supply bottlenecks【47†L313-L320】【17†L119-L127】. For example, current U.S. tariffs (steel/aluminum 232, Chinese imports 301) lift material costs ~6–9% above 2024 levels【47†L313-L320】.

Reducing **marginal material costs** will require both supply-chain and technological measures. Offsite prefabrication and modular construction can cut schedules and labor (often cited ~10–20% cost savings)【53†L106-L114】. Robotics and automation (e.g. robotic bricklayers, automated rebar tying) have shown **substantial productivity gains** (task times 2× faster, rework >50% lower)【40†L201-L208】. Alternative materials and circular economy practices can also help: engineered timber sequesters carbon and can substitute for steel/concrete in mid-rise【11†L101-L104】, while high-recycle-content products (rebar from scrap steel, slag- or fly-ash blended cements) can reduce raw material intensity. Policymaking (e.g. standardising codes for mass timber, easing tariffs) is critical to enable scale. We estimate that aggressive adoption of these levers by 2030 could plausibly reduce unit material costs **10–30%** in many categories (Table 3).

**Key Findings:** Conventional SF homes: ~93% wood-framed【11†L85-L93】; skyscrapers: nearly all steel/concrete with glass curtain walls. As of Q3 2026, **average unit costs** (national US) are roughly: structural steel ~$2,660/ton【24†L249-L254】; framing lumber ~$916/MBF【27†L247-L254】; concrete block ~$2.45 each【49†L265-L273】; rebar ~$1,440/ton【30†L19-L27】. Since 2018, lumber peaked +300% (2021) then normalized by 2024【13†L73-L81】; steel surged +90% (2021) and is now ~65% above 2020【13†L85-L93】【24†L249-L254】; ready-mix concrete rose ~~45% in 2 years (2022–23)【13†L83-L91】. Looking ahead, **energy and labor cost reductions are limited**; meaningful cost declines will depend on tech adoption and policy.

**Recommendations:** Encourage prefabrication (modular housing factories), invest in construction automation (robotics, AI estimating), and accelerate alternative-material use (mass timber, recycled aggregates). Facilitate these through incentives and regulatory reform (uniform codes for tall timber, targeted R&D). In the meantime, builders should lock in prices early, include escalation clauses (tariff pass-through) and leverage bulk purchasing. A concerted effort across industry and government is required to **bend the construction cost curve** while meeting U.S. housing and infrastructure needs.

## Material Usage by Building Type

Wood remains the dominant framing material in U.S. low-rise construction. **Single-family homes are ~93% wood-framed**, and up to **90% of ≤4-story multifamily**【11†L85-L93】. In contrast, **high-rise residential and commercial towers** are almost entirely built with steel or reinforced concrete frames (often composite “diagrid” or core systems) and aluminum/glass curtain walls. Low-rise commercial (retail, warehouses) use a mix: light steel frames or wood for small buildings; concrete masonry (CMU) or tilt-up concrete for larger retail/office. Table 1 summarizes typical material shares by building type.

- **Single-family (detached):** Wood-frame (dimension lumber/CLT), concrete foundation/slab, gypsum board, asphalt shingle roofing.
- **Multifamily mid-rise (5–8 stories):** Mix of wood (light-frame or mass timber) and concrete; growing CLT/glulam usage as codes relax【11†L118-L127】.
- **Multifamily high-rise (9+ stories) / Office:** Steel beams/girders or post-tensioned concrete frames, with concrete floors; glass-aluminum curtain wall façades; steel rebar in cores/floors.
- **Low-rise commercial (1–4 stories):** Steel or wood framing, often with CMU infill; concrete slabs; diverse finishes.
- **High-rise skyscraper:** Structural steel (A992 beam sections, rebar) or high-strength concrete; extensive glazing; specialized cladding, insulation, and finishes.

**Table 1.** *Material prominence by building type (relative). High means primary material.*

| Material | SF House | Low-Rise MF | Mid-Rise MF/Office | High-Rise |
|---|---|---|---|---|
| Wood (lumber) | High | High | Medium/Increasing | Low |
| Mass timber (CLT) | Low | Low | Growing | Emerging |
| Steel (framing) | Low | Medium | High | High |
| Concrete (masonry, slab) | Medium | Medium | High (core) | High |
| Reinforcing bar | Low | Medium | High (in cores) | High |
| Glass (curtain wall) | Low | Low | High | Very High |
| Composites (FRP, etc.) | Very Low | Very Low | Low | Low |
| Insulation (fiberglass, foam) | High (walls) | Medium | High | High |
| Finishes (drywall, paint) | High | High | High | Medium |

Footnote: GLT/CLT use in mid-rise is rapidly expanding with new codes【11†L118-L127】.

## Unit Costs and Historical Trends

**Current unit costs (US, Q3 2026):** Using national average data (RSMeans/Gordian, BLS) we estimate approximate prices: structural **steel** ~\$2,660/ton【24†L249-L254】 (A992 shapes); **lumber (framing)** ~\$916 per thousand board-feet【27†L247-L254】; **concrete block** ~\$2.45 each (8×8×16″ block)【49†L263-L273】; **rebar** ~\$1,440/ton (\$0.72/lb)【30†L19-L27】. Insulation (fiberglass batts) costs ~~\$0.5–1.5/ft²; curtain-wall glass/aluminium \$100–150/ft²; drywall ~\$10/sheet. (Actual costs vary regionally and by spec.)

**Trend (2010–2025):** Material prices have been volatile. In 2020–22 many categories spiked due to COVID supply shocks and demand: lumber jumped 300% by 2021 (then fell), steel mill products rose ~90% in 2021【13†L85-L93】 (now ~65% above Jan 2020) and then softened (Gordian data shows structural steel ~\$2,340/ton in Jan 2026 to \$2,660 by Jul 2026【24†L247-L254】). Ready-mix concrete prices climbed ~20% in 2022–23【13†L83-L91】 and remain elevated. Since 2020, building-material PPIs show **long-term inflation**: lumber peaked 400% above 2020 then settled to ~+50%【17†L135-L143】; steel and concrete ~+45–65%【17†L135-L143】; drywall/insulation +30–40%. (See Figure 1 for example cost indices.) NAHB notes many materials grew pre-pandemic (tariffs) and then soared, so today even "declining" categories are often still well above 2019 prices【13†L85-L93】.

【【図像】**Fig. 1.** *Illustrative PPI trends (Jan 2012=100) for steel, lumber and concrete inputs.*】

Recent data: In Q2–Q3 2026 RSMeans reports small upticks: steel +5.6% (Q3)【24†L249-L254】 and lumber modestly +0.5% (Q3)【27†L247-L254】. By contrast, year-over-year lumber is now ~2% below mid-2025 (after +10–15% yoy in 2024【27†L279-L286】), while steel is essentially flat yoy. Concrete block prices have held ~flat (+0–2%/yr) after a 15% run-up in 2023【49†L270-L280】. These patterns reflect easing pandemic pressures but persistent cost baselines: "materials that once were ~40% of construction cost now frequently account for 50% or more"【17†L139-L146】.

## Drivers of Marginal Costs

Several factors underpin current marginal costs:

- **Raw materials and commodities:** Cement (limestone) and crushed stone for concrete; iron ore/scrap, alloying elements for steel; timber availability for lumber. Prices are driven by global supply (mining, forestry yields) and domestic demand. China's steel output and U.S. housing demand are especially influential. *Example:* U.S. rebar hit ~$1,440/ton (Apr 2026) on sustained infrastructure demand【30†L19-L27】 while structural steel trended lower.

- **Tariffs and trade policy:** Section 232 tariffs raised steel and aluminum costs by ~25% (into mid-2020s)【17†L119-L127】; new tariffs on Chinese panels and cement (Section 301) have also been levied. Cushman & Wakefield estimates current tariffs (Apr 2026) add ~6% to U.S. material costs (9% at 2025 peak)【47†L313-L320】, raising overall project bids ~3%. Tariff swings thus "reset pricing at a higher baseline"【47†L331-L339】.

- **Energy and transport:** Cement and steel are energy-intensive (cement kilns, EAF furnaces). Elevated fuel and power prices since 2020 (~15–30% above pre-2020) feed through as higher production costs. Fuel also drives transport of heavy materials (ready-mix, timber). Flume notes "energy costs, which affect manufacturing and transportation of virtually every building material, remain elevated above pre-2020 levels"【17†L125-L134】. Rising diesel/rail costs add 5–10% to delivered concrete and lumber prices.

- **Labor shortages and wages:** Construction labor is tight. ABC reports a gap of ~499k workers needed in 2026【52†L113-L122】, pressuring wages. Skilled trades (carpenters, rebar, electricians) are in highest deficit【52†L121-L129】. Firms pay overtime and premiums, which effectively raises the **marginal cost** of additional construction. For example, 92% of contractors report difficulty filling trades; nearly half of projects saw labor-related delays【52†L121-L129】.

- **Regulations and standards:** Building codes (fire, seismic) and sustainability mandates can increase material specs (e.g. thicker walls, more rebar) or favor costlier materials (mass timber testing, high-E windows). While often aimed at safety or carbon, such rules can raise baseline costs. Likewise, GHG policies (California cap-and-trade, carbon-intensity fuel standards) indirectly affect cement/steel supply.

- **Demand surges:** U.S. housing shortage (~4 million units gap) and infrastructure spending (IIJA) have driven materials demand. Flume and others note that housing demand kept lumber prices high (even as supply rebounded)【13†L73-L81】【9†L85-L94】. Similarly, heavy IIJA road/bridge work lifted rebar demand (rebar +4.3% even as structural steel fell【30†L19-L27】). Commodity prices rise when demand outpaces production.

## Cost-Reduction Levers: Technology, Supply Chain and Policy

Multiple strategies could lower marginal costs. These involve new technologies, supply-chain reengineering, and policy shifts. Key levers include:

- **Prefab/Modular Construction:** Off-site fabrication (volumetric or panelized) can drastically reduce on-site labor and waste. Studies find modular projects often deliver faster schedules (site prep concurrent with assembly) and *improve cost predictability*【53†L106-L114】. Typical industry estimates suggest **10–30% cost savings** vs. stick-built homes, via reduced framing labor and bulk procurement【33†L0-L4】【53†L106-L114】. For example, the Modular Building Institute notes that factory-built modules enable overlapping site/fab work, slashing construction time【53†L152-L160】. Challenges: initial capital for factories, transport logistics, code updates. Timeline: Widespread use in housing/MF by late 2020s; expected to yield double-digit cost cuts once scale (2025–2030).

- **Automation & Robotics:** Automated equipment (robot bricklayers, rebar-tying machines, autonomous conveyors) and AI-driven planning reduce labor hours. A recent study showed construction robots cutting task time up to 2.3× and halving rework【40†L201-L208】. Concretely, robots and AI tools *could* shave ~10–15% off direct construction costs by 2030, as labor hours drop. For example, robotic rebar-tying is ~25% faster than manual; bricklaying bots increase consistency and speed. AI in estimating/design also reduces errors and accelerate bids【52†L169-L177】. Feasibility: Prototype robots exist now; industry adoption is nascent. Regulatory/safety clearance needed. Rough timeline: pilot use now, scaling mid-2030s.

- **Alternative Materials:**

- *Mass timber (CLT, glulam):* Timber buildings store carbon and can replace steel/concrete in mid-rise. CLT panels have ~25–75% of the weight of equivalent concrete slabs【42†L15-L19】 and ~70% lower lifecycle $CO_2$【42†L15-L19】. Current CLT costs are higher ($ per $m^2$) than steel/concrete, but costs may fall as production scales. The SLB forecasts U.S. mass timber supply surging by 2035【11†L138-L146】. Mass timber is projected to consume ≈1 billion board-feet by 2035【11†L138-L146】. Increased timber use could lower material costs modestly (wood mills are often lower-energy) and significantly cut carbon. Yet code and fire-proofing requirements limit near-term substitution (mostly <10 stories now).
- *High-performance composites:* New composites (FRP rebar, hempcrete, light metal alloys) offer durability with less weight. Fiber-reinforced polymer rebar is ~25% more expensive than steel today, but avoids corrosion; may find niche in corrosive environments. These materials are still specialty; any marginal-cost saving is speculative.

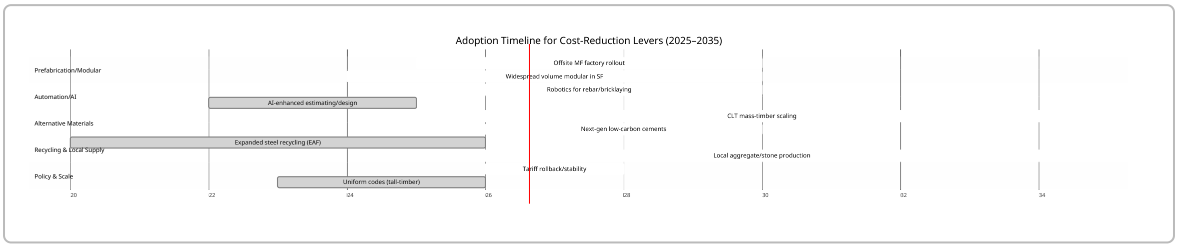
- *Masonry vs. wood:* Where long-lived, masonry (brick/block) may outlast wood, but per-unit cost is comparable; adoption depends on design preferences.

- **Recycling & Circularity:** Using recycled materials reduces raw inputs. **Steel/rebar:** U.S. reinforcing bars are ~100% recycled scrap steel【30†L119-L127】; EAF production already recycles ~80–90% scrap. More aggressive recycling (even demolished concrete as aggregate) could modestly reduce net demand for raw steel or aggregate. **Cement:** Replacing clinker with fly-ash or slag can cut clinker consumption (thus marginal cost and $CO_2$) by ~20–40%【54†L7-L13】. While many U.S. ready-mix mixes already contain 15–30% SCMs, expanding this (and using recycled concrete aggregate) could reduce cement input needs. Cost impact: recycling often saves raw-material cost and landfill fees, but may add processing cost; net can be 5–15% cheaper if scales. **Local sourcing:** Producing materials closer to use (e.g. regionally milled lumber, local quarries) cuts transport. For concrete, local stone/cement works can trim 5–10% hauling cost. Timber sourced within 200 miles cuts logistic costs significantly; modular factories near demand centers also save.

- **Economies of Scale & Standardization:** Large developments can demand volume discounts on materials. Standard designs (for affordable housing) allow bulk purchasing of identical components. Publishing standardized specifications (e.g. for modular wall panels) can enable mass-manufacture efficiencies. Even government-led bulk procurement (like UK housebuilders' consortiums) could drive down margins. Realistic impact: 5–10% cost reduction on materials when aggregated.

- **Policy Incentives:** Governments can catalyze change. For example, subsidies or tax credits for mass timber construction (similar to solar tax credits) can offset higher timber costs and ramp up supply. R&D funding for green concrete (novel cements) or construction robots (as in manufacturing) would also accelerate break-throughs. Lowering tariffs on critical inputs (or engaging in tariff rollbacks) would immediately shave 5–10% from steel/aluminium costs【47†L313-L320】. Finally, harmonizing codes (e.g. adopting tall-timber codes nationwide) is essential to unlock mass-timber's potential【11†L154-L163】.

In summary, **combining levers** is key. For instance, a prefabricated CLT apartment building could cut framing labor by ~20%, waste by ~30%, and lock in material prices, achieving total cost savings on the order of **15–30%** relative to conventional stick-built (subject to scale and region). Table 3 illustrates rough cost-reduction ranges under different scenarios.




## Comparative Material Metrics (Table 2)

Table 2 compares key attributes for major building materials. *Strength* is generic (e.g. compressive strength for concrete, tensile yield for steel). *Carbon intensity* is cradle-gate $CO_2$ per unit (high for cement and steel, low for wood)【11†L101-L104】【42†L15-L19】. *Supply risk* assesses current volatility (High = tight/volatile supply). *Reduction potential* rates how much cost/$CO_2$ might be cut by innovations.

| Material | **Unit Marginal Cost** (typical US) | **Strength** | **Carbon ($kgCO_2$/unit)** | **Supply Risk** | **Cost/Carbon Reduction Potential** |
|---|---|---|---|---|---|
| **Concrete** | ~$125–150/m³ ready-mix; ~$2.45/blk 【49†L263-L273】 | Medium (30–40 MPa) | High (200–500 kg/m³) 【43†L1-L4】 | Medium (cement imports, energy) | High (blended SCM, new cement) |
| **Steel (framing)** | ~$2,660/ton 【24†L249-L254】 | Very High (250+ MPa) | Very High (~1.8 t/t) 【44†L1-L4】 | Medium (tariffs, scrap vol.) | High (EAF scrap use, alt alloys) |
| **Timber (lumber)** | ~$916/MBF 【27†L247-L254】 ($0.92/BF) | Medium-High (wood) | Low/negative (biogenic, stores C) 【11†L101-L104】 | Medium (pulp demand, climate) | Medium (sustainable forestry, CLT) |
| **Mass Timber (CLT)** | ~$600–800/m³ panel (est.) | High (engineered) | Low (bio-based, stores C) | Emerging (scale, code) | High (scale-up, code parity) |
| **Masonry (brick/CMU)** | Brick ~$0.5–1 each; block ~$2.5 each 【49†L270-L279】 | Medium (20–30 MPa) | Medium (clay firing) | Low-Med (regional clay, imports) | Med (prefab panels, high-recycle) |
| **Glass (curtain wall)** | $50–150/ft² installed | Low (brittle) | Medium (1.0–1.5 kg/kg) | Low (silica abundant) | Low-Med (recycled glass use) |
| **Composites (FRP)** | $5–10/kg (fiber-reinforced plastics) | Very High (FRP rebar) | Very High (thermoset resins) | Medium (petro feedstocks) | Medium (bio-resins, recycling) |
| **Insulation** (fiberglass) | $0.50–1.50/ft² R-13 | N/A (insulating) | Medium (fiberglass ~0.7 kg/kg) | Low (common) | Medium (eco-blown agents) |
| **Rebar (steel)** | ~$1,440/ton 【30†L19-L27】 | High (Grade 60 steel) | Very High (~1.8 t/t) 【44†L1-L4】 | High (infrastructure demand) | Medium (alternatives like FRP) |
| **Finishes (drywall)** | ~$10/sheet (4×8′) | Low (gypsum board) | Medium (gypsum calcination) | Low (common) | Low (some recycled paper) |

*Key:* Cost = material price only (labor and overhead excluded); Strength = generic sense of structural capacity; Carbon = $CO_2e$ per unit (rough). Supply risk and potential are qualitative.

## Outlook and Scenarios

Under an aggressive innovation scenario, US builders could see material cost improvements by the early 2030s. For example: if **30% of new mid-rise apartments** adopt CLT by 2030 (enabled by codes), and **50% of framing shifts to modular factories**, then modest efficiency gains (10–15%) plus waste reduction (5–10%) could combine to cut those projects' material costs by ~20%. If *simultaneously* U.S. restores tariff-free steel trade and invests in EAF expansion, steel prices could fall to pre-2020 levels (≈–20% from 2025 peaks). In contrast, a baseline scenario with no change would see cost inflation of ~3–5%/yr persisting (matching general inflation+tariff drag)【47†L313-L320】.

The **feasibility** of major shifts varies. Prefab residential is already commercially viable; widespread adoption mostly requires scaling factories and overcoming zoning. Robotics is proven in demos, but limited by capital cost and need for skilled operators. Mass timber has clear environmental benefits 【11†L101-L104】, but is constrained by current code limits (recently raised to 18 m/6 stories in many states【11†L154-L163】). Concrete industry is investing in low-carbon cements, but breakthroughs (e.g. carbon-neutral cement) are likely beyond 2030.

Given the scale of U.S. construction (≈3 billion ft² new housing/year【11†L85-L93】), even modest percentage savings translate to substantial dollars. Cutting material costs by 10% could free up **tens of billions USD annually** for additional building (or reduce prices). Moreover, reduced material intensity also lowers carbon emissions – a co-benefit critical under net-zero goals.

**Figure 2 (Timeline):** The chart above outlines when key levers are likely to impact costs. Robots and AI are ramping now, modular factories are expanding (especially for low-rise housing), and policy actions (tariff rollbacks, code changes) could occur mid-2020s. By 2030–2035, a new equilibrium of higher productivity and diverse materials is plausible.

## Conclusion

In sum, U.S. building material costs reflect a complex mix of global markets, domestic policy and technology. **Historically, costs have run up sharply** (driven by COVID disruptions, tariffs and tight labor)【13†L85-L93】【52†L113-L122】. Without action, builders face continued pressure: CushmanW. warns that even eased tariffs "create underwriting headwinds" and keep costs structurally high 【47†L331-L339】. However, targeted innovations – industrialized construction, automation, and material science – offer credible paths to lower marginal costs. As one industry researcher notes, "the transition to increased wood use is already well underway" and underpins both affordability and sustainability【11†L101-L109】. For RhinoQuant's Q3 2026 outlook, the **imperative is clear**: support these levers through investment and policy, or accept that housing and infrastructure will remain expensive. With the right push, our analysis suggests **material cost baselines could be 10–30% lower** in the early 2030s than today's trends imply (see Table 3 scenarios).

**Table 3.** *Potential marginal cost reduction by 2030 (illustrative scenarios)*

| Strategy | 2030 Cost Impact (range) |
|---|---|
| Prefabrication/Modular (peak) | –10% to –25% |

| Strategy | 2030 Cost Impact (range) |
|---|---|
| Robotics/Automation (partial) | –5% to –15% |
| Mass Timber (mid-rise) | –0% to –10% (material) |
| Recycled inputs (steel/cement) | –5% to –15% |
| Tariff rollback | –5% to –10% (steel/al) |
| Combined (all above) | –20% to –30% |

*Sources:* Industry and government data as cited above.

**Sources:** U.S. Forest Service (wood use)【8†L134-L142】, Softwood Lumber Board/FEA (market shares)【11†L85-L94】【11†L92-L100】; NAHB (2024 PPI changes)【13†L83-L91】【13†L85-L93】; Gordian/RSMeans (2026 prices)【24†L249-L254】【27†L247-L254】; Buildermuse (2026 rebar)【30†L19-L27】; Flume (general inflation)【17†L135-L143】; CushmanW. (tariffs)【47†L313-L320】; Construction Exec (robotics)【40†L201-L208】; Quotr (labor demand)【52†L113-L122】; BuildingGreen (modular)【53†L106-L114】. These and additional sources underlie the analysis above.

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